

The Output Gap Does Not Exist

What “the gap” measures depends on the model—and the data

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October 27, 2025

Main Point

- ▶ There is no *the* output gap
- ▶ Empirical output-gap measures capture different objects.
- ▶ Estimates vary widely; interpretation is unclear.
- ▶ Fjern slide?

Conventional setup: state space with a single gap

Output gap estimates split observed output into a trend (potential output) and a stationary cycle (gap):

$$y_t = \tau_t + c_t + \varepsilon_t^y.$$

(τ_t, c_t) are unobserved states:

$$\begin{aligned} s_t &= \Phi s_{t-1} + \xi_t, & \xi_t &\sim \mathcal{N}(0, \Sigma_\xi), \\ x_t &= Hs_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \Sigma_\varepsilon). \end{aligned}$$

s_t is the state vector (e.g. $s_t = (c_t, \tau_t, \Psi_t, \mu_t, \dots)$). x_t are observables

$$x_t = \begin{bmatrix} y_t \\ z_t \end{bmatrix}, \quad z_t = (u_t, \pi_t, E_t \pi_{t+1}, \dots).$$

Covers HP/BP filters, BN, BQ, UC models, PF & SVARs.

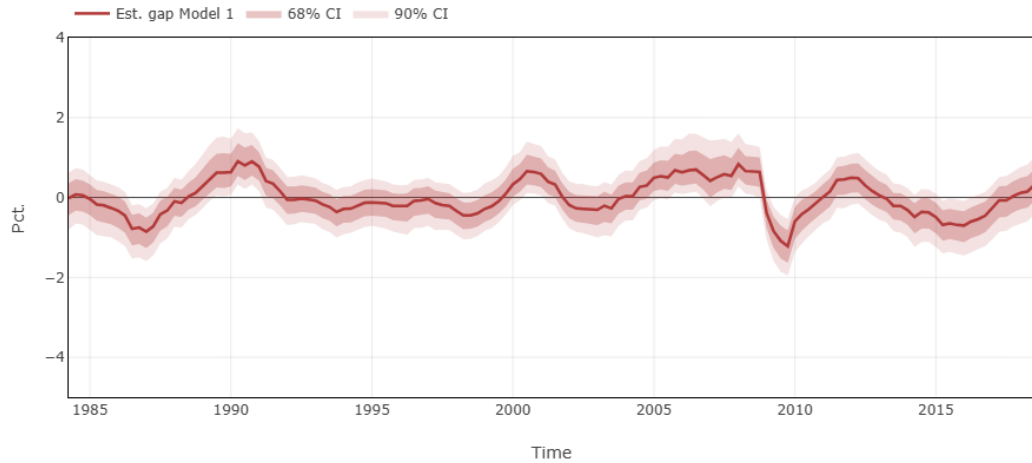
Example. Model 1: *Inflation Model*

- ▶ Idea: the output gap links real activity to inflation/expectations.
- ▶ One possible setup (common states + idiosyncratic drifts/noise):

$$\begin{pmatrix} y_t \\ \pi_t \\ E_t \pi_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ \delta_\pi & 1 \\ \delta_{E\pi} & 1 \end{pmatrix}}_{\text{common}} \begin{pmatrix} c_t \\ \mu_t^\pi \end{pmatrix} + \underbrace{\begin{pmatrix} \tau_t \\ 0 \\ \mu_t^{E\pi} \end{pmatrix}}_{\text{drifts}} + \underbrace{\begin{pmatrix} \psi_t^y \\ \psi_t^\pi \\ \psi_t^{E\pi} \end{pmatrix}}_{\text{noise}}.$$

- ▶ c_t is the "gap", τ_t is the "potential".
- ▶ μ_t^π is trend inflation capturing long term inflation expectations (steady state inflation). $\mu_t^{E\pi}$ is potential long term expectation bias
- ▶ ψ are idiosyncratic cycles to capture unexplained transitory fluctuations

Model 1: Estimated Gap



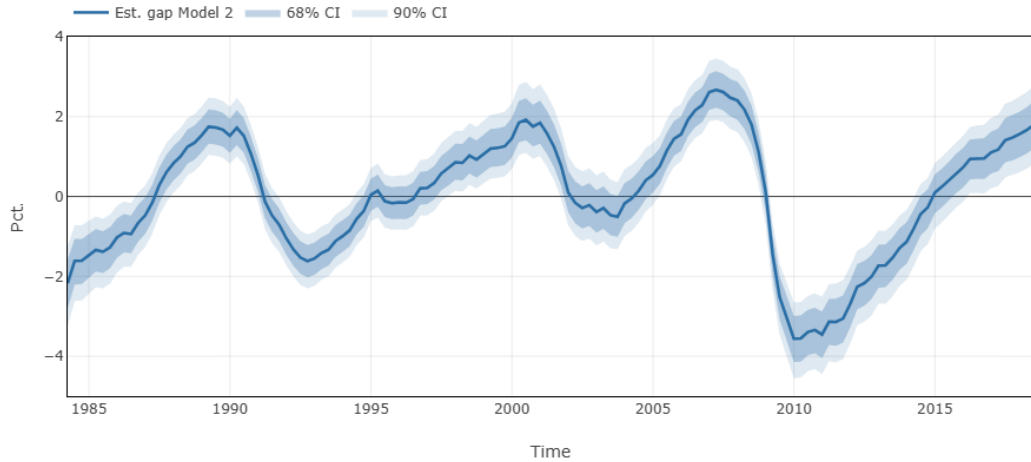
Example. Model 2: *Inflation + Labour*

- Idea: the gap drives *broader* cyclical comovement: We include labour.

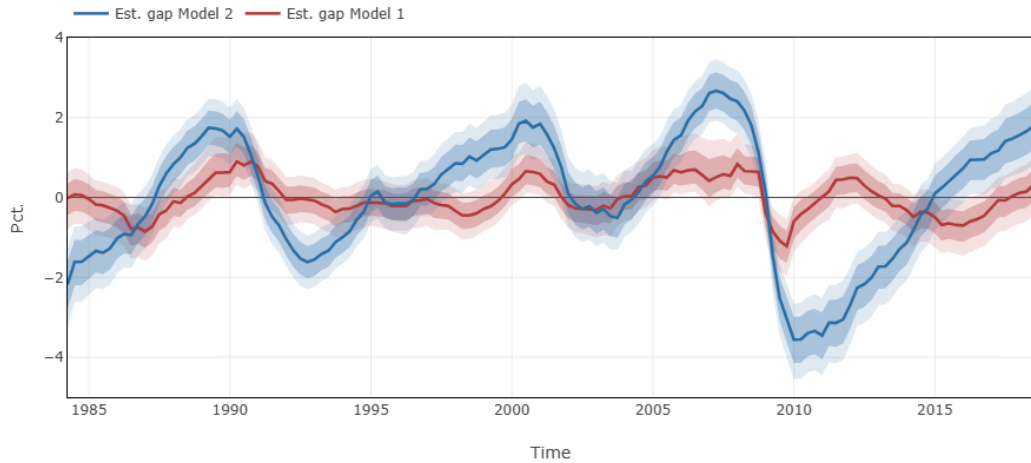
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- Model otherwise unchanged

Model 2: Estimated Gap



Estimated Gaps



They are not the same? So which gap is it then?

- ▶ The two models deliver *very different* gaps on the same data.
- ▶ Which one is "right"? Not at all obvious
 - ▶ Different because they identify *different* gaps?
 - ▶ Or no gap at all?

Critique: ALL conventional methods assume *one gap*

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- ▶ *But:* if more than one economically distinct gap exists, identification and economic interpretation becomes fragile.

Multiple *Gaps* Exist

- ▶ **Nominal rigidities:** slow price adjustment $\Rightarrow$ demand–supply imbalances show up in inflation.
- ▶ **Real rigidities:** market power, matching frictions, etc. distort allocation even with flexible prices.
- ▶ Implication: policy-relevant deviations from (i) efficient allocation and (ii) flex-price allocation co-exist.
- ▶ What gaps did we identify? We need a structured framework to think about this

Plan

The output gap does not exist because of multiple distinct frictions and distortions

- ▶ This is a general insight across many macro models
- ▶ Also true in the basic/stripped-down version of the workhorse model in modern macro: the NK model

What we do:

- ▶ Use a stripped-down NK model as DGP.
- ▶ Re-estimate naive models and compare to **true** efficient and flex-price gaps.

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 3. Cost-push shocks $\Rightarrow$ shocks to firm markups

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- ▶ Proportional if no cost-push shocks. Coincide if efficient steady state

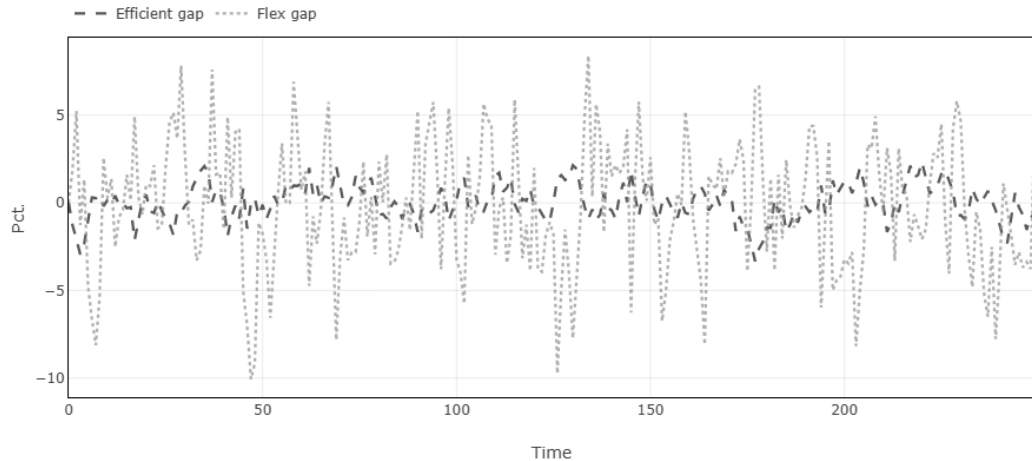
NK-model: Why do we want to know the gaps?

- ▶ *Actual output* $\neq$ *Efficient output* $\Rightarrow$ output below productive capacity of economy
- ▶ *Actual output* $\neq$ *Flexible-price output* $\Rightarrow$ aggregate demand $\neq$ firm optimal supply
 $\Rightarrow$ firms update prices when allowed $\Rightarrow$ Inflation/deflation
- ▶ In NK model ***efficient output gap*** and ***inflation*** enter social loss function

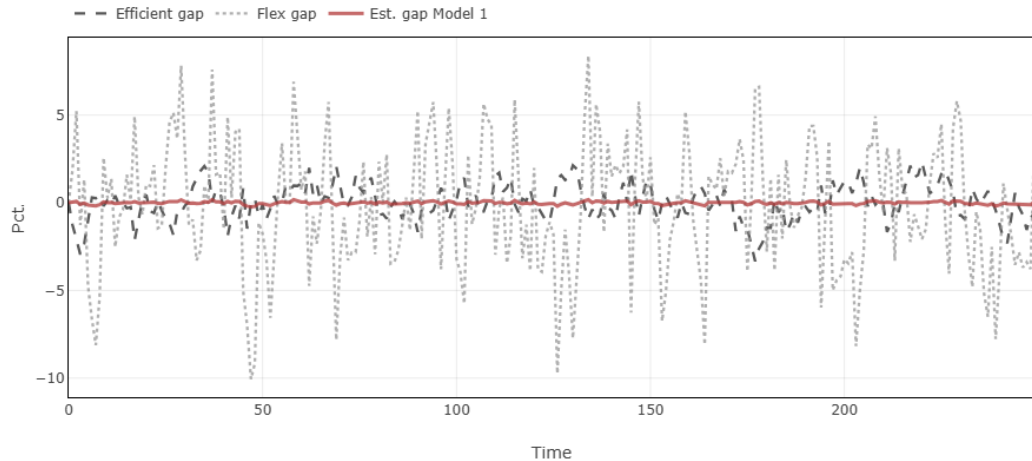
DGP and Simulation Setup

- ▶ Simulate 250 quarters of data from textbook NK model with parameter calibration following Galí (2015)
- ▶ Estimate the two *naive* empirical models on the simulated data.
- ▶ Compare estimated gaps against true c_t^{eff} and c_t^{flex} .

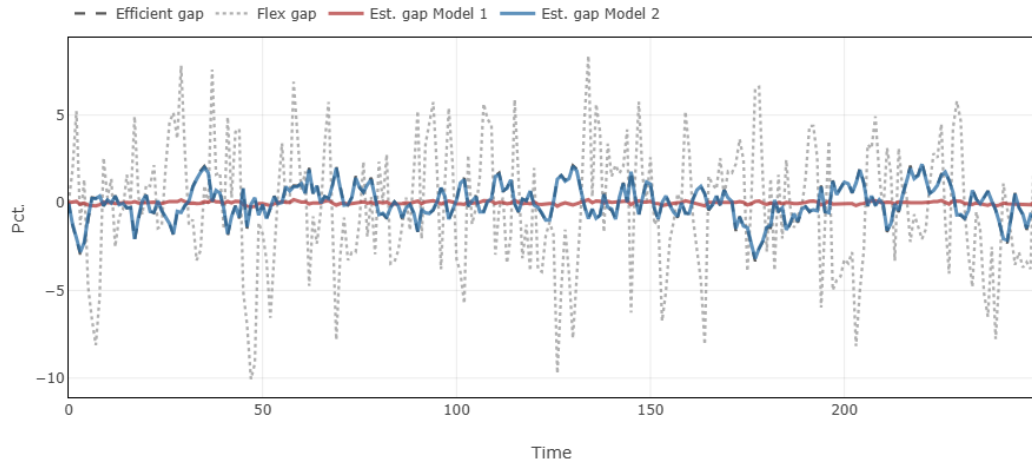
Results. First Simulation



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Results. Monte Carlo Summary

Table: Identification metrics

DGP	Metric	Model 1		Model 2	
		Eff	Flex	Eff	Flex
NK	MAE	0.759 (0.041)	2.752 (0.203)	0.101 (0.204)	3.147 (0.219)
	Corr	-0.145 (0.076)	0.967 (0.007)	0.925 (0.237)	-0.444 (0.200)

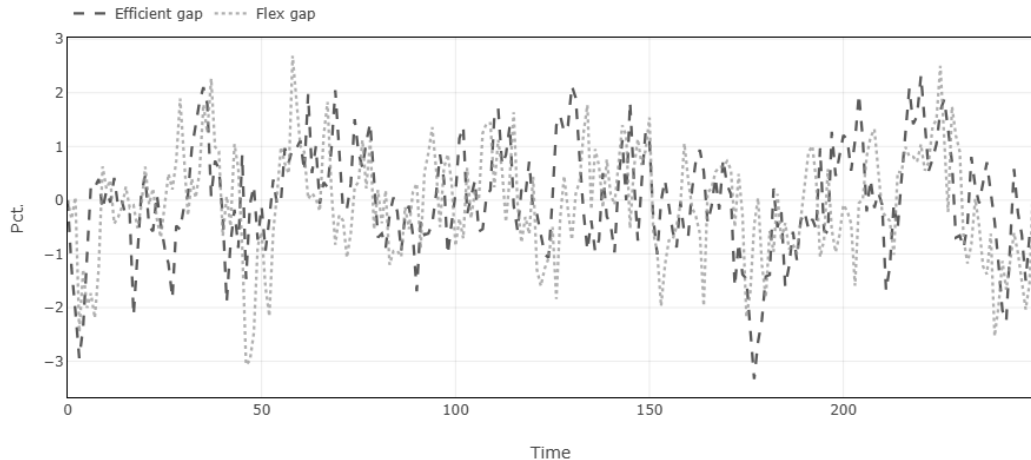
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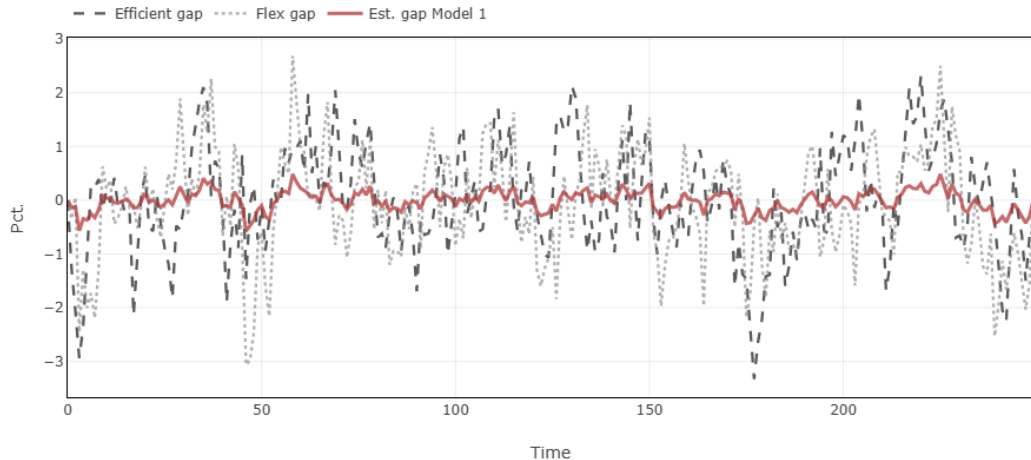
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- It would seem model 2 is consistently good for *efficient gap* while *Flex gap* is not recovered

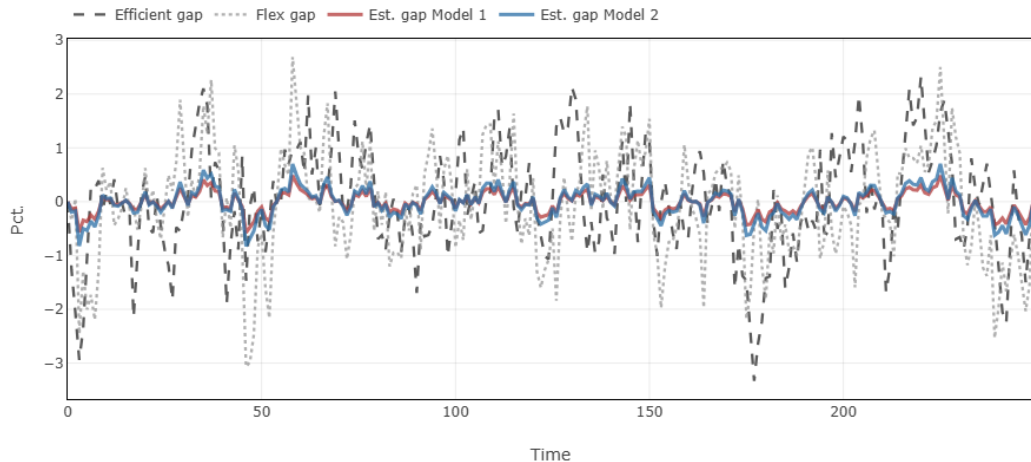
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NK steep	MAE	0.623 (0.024)	0.506 (0.063)	0.470 (0.273)	0.662 (0.250)
	Corr	0.558 (0.077)	0.950 (0.006)	0.613 (0.408)	0.551 (0.629)

- Model 2 doesn't recover the efficient gap any longer. Changing one parameter in DGP changes the extracted gap.

The "Naive" Empirical Models

- ▶ Hammer home point about issues with empirical output gap methods. The researcher does not choose which gap is being estimated if any. The data does. And it does not tell the researcher
- ▶ Understreg at vores indsigt er generel og ælder for alle gænmgde metoder (inkl. filtre osv) der anvendes. Nogle metoder er bedre end andre, ja, men alle er dårlige fordi *de antager ét gab*.

What can we do? Identifying gaps in the NK model

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Okun's law (real side)

$$y_t = a_t + (1 - \alpha)n_t, \quad \tau_t^{\text{eff}} = a_t + (1 - \alpha)n_t^{\text{eff}},$$

$$n_t^{\text{eff}} = \frac{1 - \sigma}{\sigma(1 - \alpha) + \varphi + \alpha} a_t.$$

- ▶ Efficient gap: $c_t^{\text{eff}} = (1 - \alpha)(n_t - n_t^{\text{eff}})$.
- ▶ If $\sigma = 1$: $n_t^{\text{eff}} = 0$. Else $n_t^{\text{eff}} \approx 0$ for reasonable parameters $\Rightarrow$ hours tightly track c_t^{eff} proportionally.

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Phillips curve (nominal link)

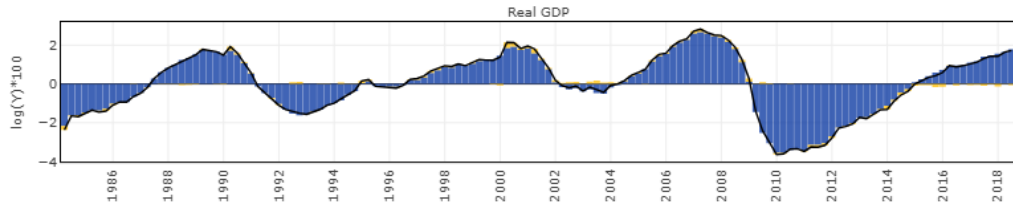
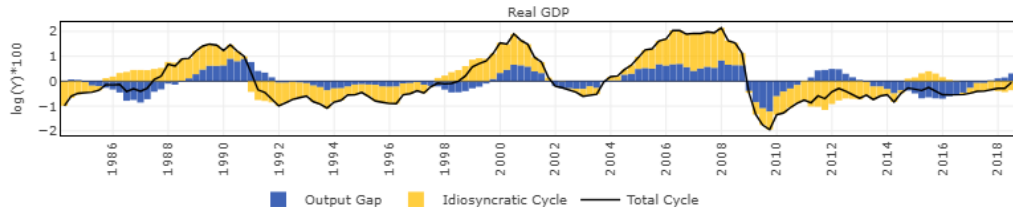
$$\pi_t - \beta E_t \pi_{t+1} = \kappa c_t^{\text{flex}} = \kappa c_t^{\text{eff}} + \underbrace{u_t}_{\text{cost-push}}.$$

$$u_t = \kappa (c_t^{\text{flex}} - c_t^{\text{eff}}) \Rightarrow c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t.$$

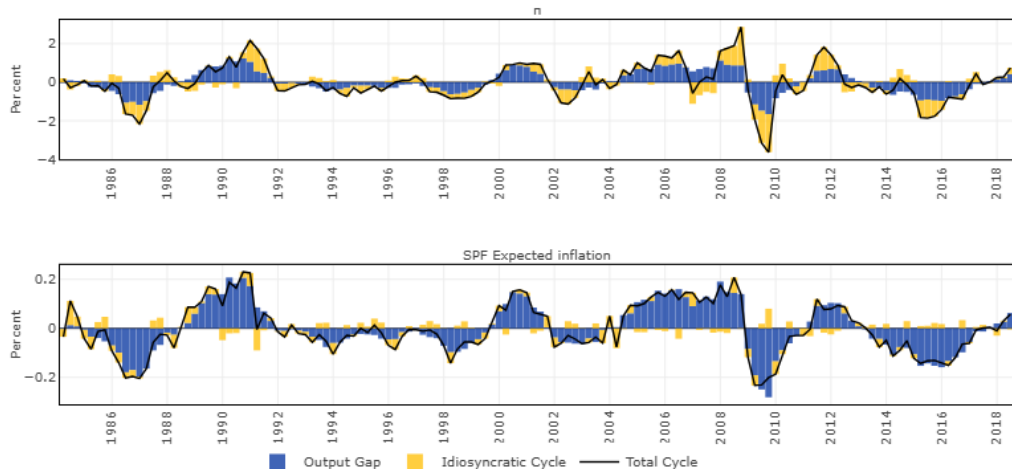
Identification in model 1 and model 2

- ▶ **Model 1 (inflation):** nests NKPC $\Rightarrow$ aims at c_t^{flex} , but weak identification in small samples/weak nominal–real comovement.
- ▶ **Model 2 (inflation+labour):** blends real and nominal blocks; target depends on the DGP.
 - ▶ If hours–output link dominates $\Rightarrow$ mimics c_t^{eff} . If nominal link dominates $\Rightarrow$ drifts toward c_t^{flex}

What did we estimate? Decomposing cyclical fluctuations



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Goal: target the NK objects

- ▶ Use NK identification to **directly** estimate c_t^{eff} (efficient gap) and u_t (cost-push).
- ▶ Recover c_t^{flex} via $c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t$.
- ▶ Keep the empirical system *minimal* and structurally informed.

Textbook NK identification

Real block (Okun-like)

$$y_t = a_t + (1 - \alpha)n_t, \quad \tau_t^{\text{eff}} = a_t + (1 - \alpha)n_t^{\text{eff}}, \quad c_t^{\text{eff}} = (1 - \alpha)(n_t - n_t^{\text{eff}}).$$

n_t^{eff} roughly constant $\Rightarrow$ hours pin down the efficient gap.

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n_t^{eff} roughly constant $\Rightarrow$ hours pin down the efficient gap.

Nominal block (NKPC)

$$\pi_t - \beta E_t \pi_{t+1} = \kappa c_t^{\text{eff}} + u_t, \quad c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t.$$

c_t^{eff} identified $\Rightarrow$ inflation identifies u_t and κ given c_t^{eff} .

The two-gap model. Measurement equations

Observables $x_t = (y_t, e_t, u_t, \pi_t, E_t\pi_{t+1})'$, states $s_t = (\tau_t^{\text{eff}}, c_t^{\text{eff}}, u_t)'$:

$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \\ E_t\pi_{t+1} \end{pmatrix}}_{x_t} = \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 0 & \delta_e & 0 \\ 0 & \delta_u & 0 \\ 0 & \delta_E + \kappa & \gamma E + 1 \\ 0 & \delta_E & \gamma E \end{pmatrix}}_H \underbrace{\begin{pmatrix} \tau_t^{\text{eff}} \\ c_t^{\text{eff}} \\ u_t \end{pmatrix}}_{s_t} + \underbrace{\begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \\ \varepsilon_t^u \\ \varepsilon_t^\pi \\ \varepsilon_t^{E\pi} \end{pmatrix}}_{\varepsilon_t}, \quad c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t.$$

Intuition: Rows 1–3 load only on the efficient cycle $(\tau^{\text{eff}}, c^{\text{eff}})$; rows 4–5 load on nominal components with δ_E, κ linking activity to inflation.

The two-gap model. State dynamics

Efficient (real) cycle c^e

$$\begin{pmatrix} c_t^e \\ c_t^{e*} \end{pmatrix} = \rho_e \begin{pmatrix} \cos \lambda_e & \sin \lambda_e \\ -\sin \lambda_e & \cos \lambda_e \end{pmatrix} \begin{pmatrix} c_{t-1}^e \\ c_{t-1}^{e*} \end{pmatrix} + \begin{pmatrix} \eta_t^e \\ \eta_t^{e*} \end{pmatrix}$$

Cost-push (nominal) cycle u

$$\begin{pmatrix} u_t \\ u_t^* \end{pmatrix} = \rho_\pi \begin{pmatrix} \cos \lambda_\pi & \sin \lambda_\pi \\ -\sin \lambda_\pi & \cos \lambda_\pi \end{pmatrix} \begin{pmatrix} u_{t-1} \\ u_{t-1}^* \end{pmatrix} + \begin{pmatrix} \eta_t^\pi \\ \eta_t^{\pi*} \end{pmatrix}$$

Cycles are general AR(2) processes where $\rho \in (0, 1)$ governs *persistence* (damping), $\lambda \in (0, \pi)$ sets the *frequency* (period $T = \frac{2\pi}{\lambda}$)

Efficient trend τ_t^{eff}

$$\tau_t^{\text{eff}} = \tau_{t-1}^{\text{eff}} + \zeta_t^\pi$$

The two-gap model. Shock covariances

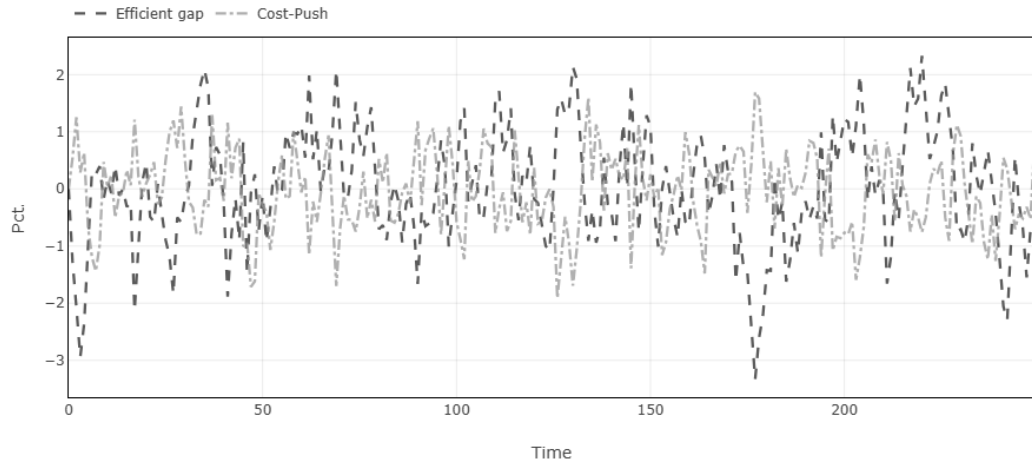
Endogenous policy response to cost-push shocks $\Rightarrow \text{Cov}(\eta_t^e, \eta_t^\pi) \neq 0$.

$$\eta_t = (\eta_t^e \quad \eta_t^{e*} \quad \eta_t^\pi \quad \eta_t^{\pi*} \quad \zeta_t^\pi)' \sim \mathcal{N}(0, Q),$$

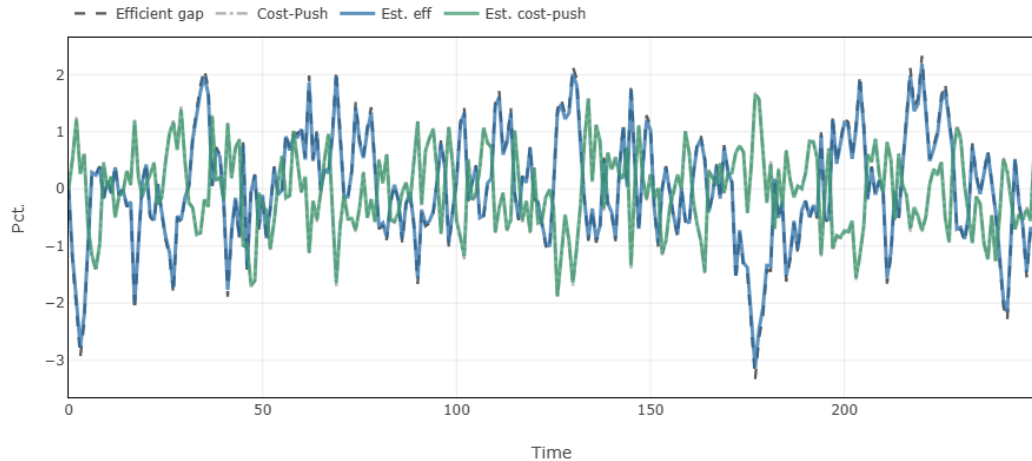
$$Q = \begin{pmatrix} Q_{e,\pi} & 0 \\ 0 & \sigma_{\zeta^\pi}^2 \end{pmatrix}, \quad Q_{e,\pi} = \begin{bmatrix} \sigma_e^2 l_2 & \rho \sigma_e \sigma_\pi l_2 \\ \rho \sigma_e \sigma_\pi l_2 & \sigma_\pi^2 l_2 \end{bmatrix}.$$

Interpretation: ρ governs **real**–**nominal** comovement through correlated innovations.

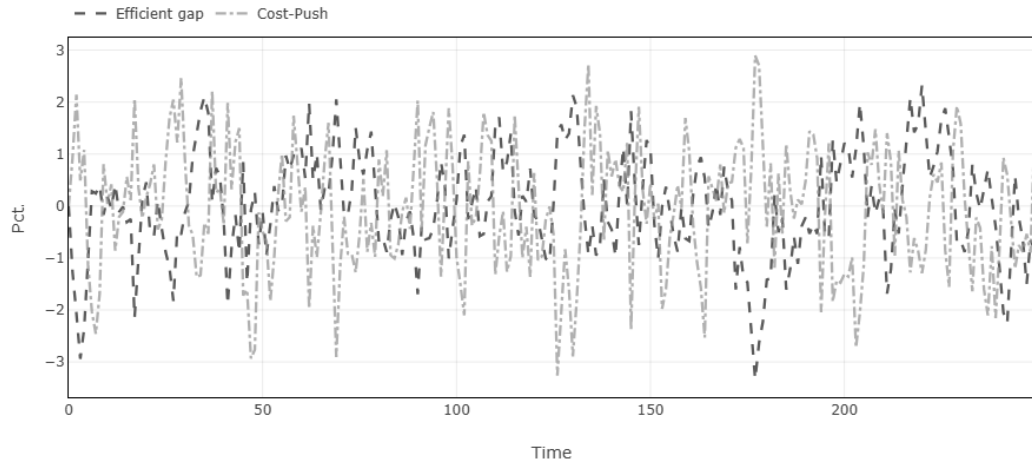
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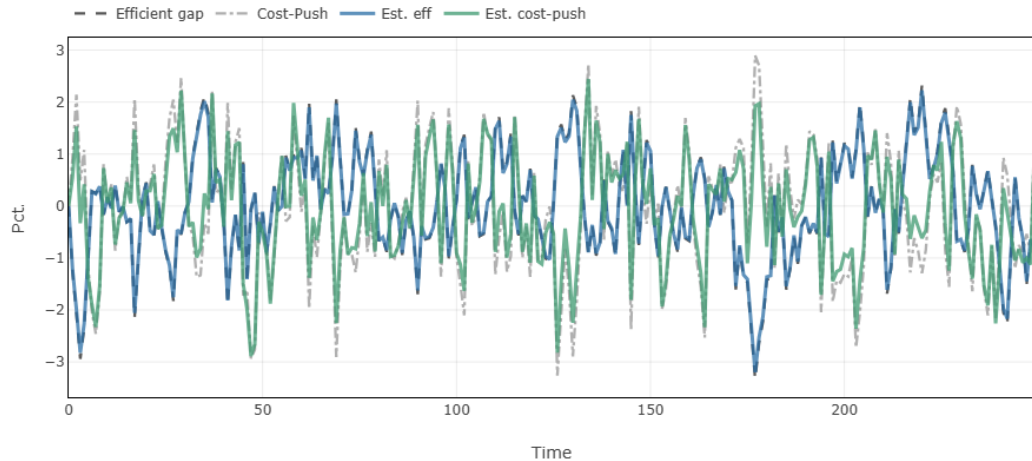
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Estimated States high NKPC slope κ



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Results: Efficient Gap and Cost-Push

Table: Identification metrics.

DGP	Metric	2-gap model	
		Eff	u
NK	MAE	0.009 (0.007)	0.101 (0.054)
	Corr	1.000 (0.000)	0.990 (0.008)
NK steep	MAE	0.013 (0.017)	0.192 (0.143)
	Corr	1.000 (0.000)	0.974 (0.031)

How about c_t^{flex} ?



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Results: NKPC Slope (κ)

Table: Estimated Phillips-curve slope (κ)

DGP(κ_{true})	2-gap model
NK (0.172)	0.299 (0.048)
NK steep (1.01)	0.838 (0.224)

Results: Efficient vs. Flex Gaps

Table: Identification metrics.

DGP	Metric	2-gap model		
		Eff	u	Flex
NK	MAE	0.009 (0.007)	0.101 (0.054)	0.971 (0.491)
	Corr	1.000 (0.000)	0.990 (0.008)	0.999 (0.000)
NK steep	MAE	0.013 (0.017)	0.192 (0.143)	0.292 (0.313)
	Corr	1.000 (0.000)	0.974 (0.031)	0.998 (0.001)

Targeting the Flex Gap: Challenges

- ▶ $c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t \Rightarrow$ needs both cycles to be recovered precisely.
- ▶ Directly modeling c_t^{flex} in a (y, π) -only system performs worse. Weak identification for small κ (holds even for fixed known κ)

Known $\kappa \neq$ Easy Flex Gap

- ▶ With known κ , one can back out c_t^{flex} from NKPC.
- ▶ But inflation/expectations noise directly contaminates the inferred flex gap.
- ▶ Better: jointly estimate c_t^{eff} and u_t ; then form $c_t^{\text{flex}} = c_t^{\text{eff}} + \kappa^{-1} u_t$ so noise is absorbed handled.

Bringing it to the data

We use insights from the simulation exercise to motivate an empirical model to take to the data

- ▶ We pin down the *efficient* gap from the *real* side of the economy
- ▶ We identify a *cost-push* process from common residuals in inflation and inflation expectations
- ▶ We back out a *flex-price* gap using NK structure

Measurement equation: Real / Okun block (1/3)

$$\underbrace{(y_t)}_{y_t^{obs}} = \underbrace{(1 \quad 0 \quad 0 \quad 0 \quad 0)}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: Output loads only on the **efficient cycle** (g_t^e, g_t^{e*}).

Measurement equation: Real / Okun block (2/3)

$$\underbrace{\begin{pmatrix} y_t \\ e_t \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \delta_{e,1} & \delta_{e,2} & 0 & 0 & 0 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: Employment co-moves with the efficient cycle via $(\delta_{e,1}, \delta_{e,2})$.

Measurement equation: Real / Okun block (3/3)

$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \delta_{e,1} & \delta_{e,2} & 0 & 0 & 0 \\ \delta_{u,1} & \delta_{u,2} & 0 & 0 & 0 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: Unemployment ties to the efficient cycle; signs in $(\delta_{u,1}, \delta_{u,2})$ encode Okun's law.

Measurement equation: **Nominal** block + NK slope (1/3)

$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} \delta_{E,1} + \kappa & \delta_{E,2} + \delta & \gamma_{E,1} + 1 & \gamma_{E,2} + \gamma & 1 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: Inflation loads on **cost-push components** and the **anchor**, with (κ, δ) bringing real activity into inflation.

Measurement equation: Nominal block (2/3)

$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \\ F_t^{uom} \pi_{t+4} \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} \delta_{E,1} + \kappa & \delta_{E,2} + \delta & \gamma_{E,1} + 1 & \gamma_{E,2} + \gamma & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: UoM expectations share the common nominal components and anchor; $\delta_{E,\cdot}$ reflect real activity loadings.

Measurement equation: Nominal block (3/3, complete)

$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \\ F_t^{uom} \pi_{t+4} \\ F_t^{spf} \pi_{t+4} \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} \delta_{E,1} + \kappa & \delta_{E,2} + \delta & \gamma_{E,1} + 1 & \gamma_{E,2} + \gamma & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t}$$

Intuition: SPF expectations mirror UoM. All δ entries (real activity loadings) are blue as requested.

Full measurement equation system (all rows)

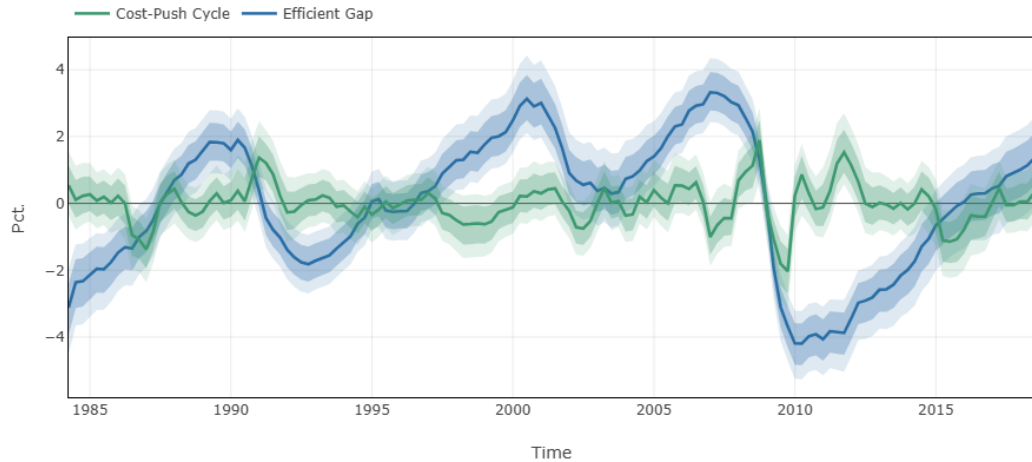
$$\underbrace{\begin{pmatrix} y_t \\ e_t \\ u_t \\ \pi_t \\ F_t^{uom} \pi_{t+4} \\ F_t^{spf} \pi_{t+4} \end{pmatrix}}_{y_t^{obs}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \delta_{e,1} & \delta_{e,2} & 0 & 0 & 0 \\ \delta_{u,1} & \delta_{u,2} & 0 & 0 & 0 \\ \delta_{E,1}+\kappa & \delta_{E,2}+\delta & \gamma_{E,1}+1 & \gamma_{E,2}+\gamma & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \\ \delta_{E,1} & \delta_{E,2} & \gamma_{E,1} & \gamma_{E,2} & 1 \end{pmatrix}}_Z \underbrace{\begin{pmatrix} g_t^e \\ g_t^{e*} \\ g_t^\pi \\ g_t^{\pi*} \\ \mu_t^\pi \end{pmatrix}}_{s_t} + \underbrace{\begin{pmatrix} \psi_t^y \\ \psi_t^e \\ \psi_t^u \\ \psi_t^\pi \\ \psi_t^{uom} \\ \psi_t^{spf} \end{pmatrix}}_{\Psi_t} + \underbrace{\begin{pmatrix} \mu_t^y \\ \mu_t^e \\ \mu_t^u \\ 0 \\ \mu_t^{uom} \\ \mu_t^{spf} \end{pmatrix}}_{\mu_t}$$

- Rows 1–3 (Real/Okun): y_t, e_t, u_t load only on (g_t^e, g_t^{e*}) .
- Rows 4–6 (Nominal): π_t and expectations load on $(g_t^\pi, g_t^{\pi*}, \mu_t^\pi)$; (κ, δ) tie real activity to inflation.

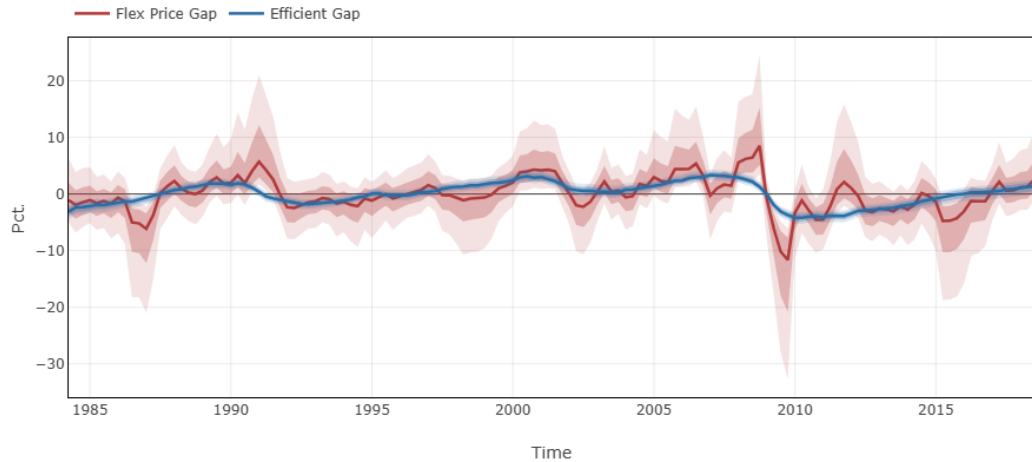
Identification intuition

- ▶ **Efficient gap g_t^e (real slack).** Identified by **co-movement in (y_t, e_t, u_t)** through the Okun block (rows 1–3). No nominal terms load here $\Rightarrow$ clean read on real slack.
- ▶ **Cost–push wedge g_t^π (nominal pressure).** Identified by **common movement in π_t and expectations** (rows 4–6) *conditional* on activity: π_t loads on **$(\kappa, \delta) \cdot (g_t^e, g_t^{e*})$** , while both π_t and $F\pi_{t+4}$ load on **$(g_t^\pi, g_t^{\pi*}, \mu_t^\pi)$** . This joint structure separates nominal pressure from real slack.
- ▶ **Takeaway.** **Real block** pins down g_t^e ; **nominal block**, together with its **co-movement with g_t^e in π_t** , pins down g_t^π and the anchor μ_t^π .

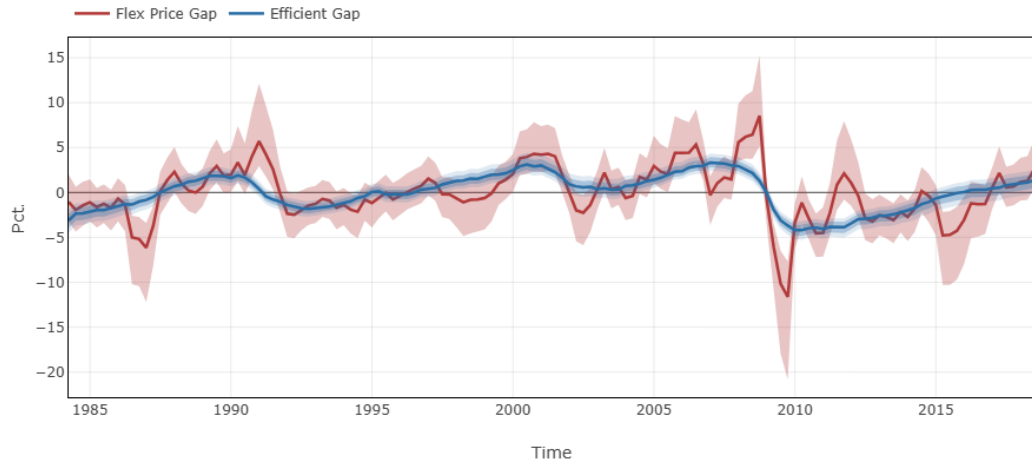
Estimated states



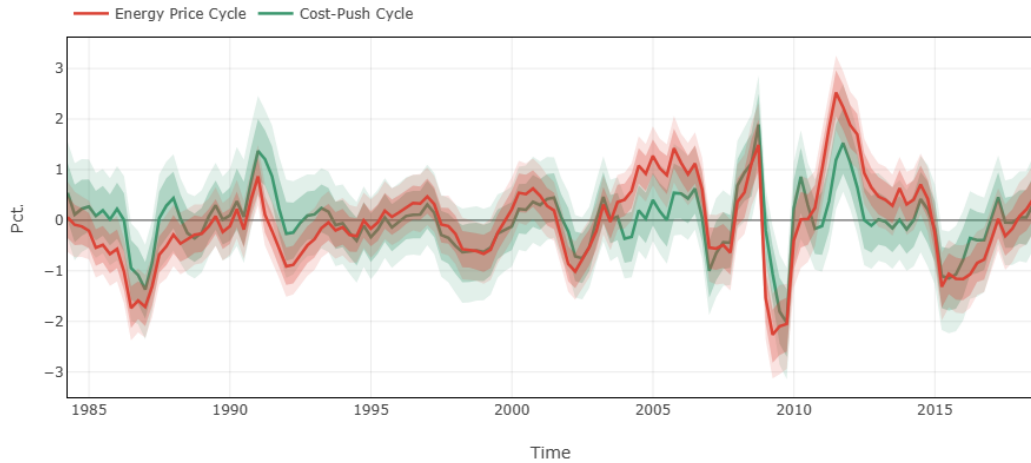
Estimated states



Estimated states



The cost-push cycle



Cycle Decompositions

Parameter Posteriors